Movie Library Application

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Problem Statement

Challenge: Efficiently managing a personal movie library requires a unified system that bridges live global metadata (TMDB) with private user data (ratings/reviews) while ensuring backend security.

Solution: A full-stack **Next.js** application that handles the complete data lifecycle:

- **Real-time Discovery:** Search movies via secure API routes.
- **Personalized CRUD:** Add, update, and delete titles in a local `db.json` database.
- **Dynamic UX:** Automated generation of media-rich detail pages for every movie.

Expected Output: A secure, data-driven library where all content is managed through a clear API structure and dynamic routing.

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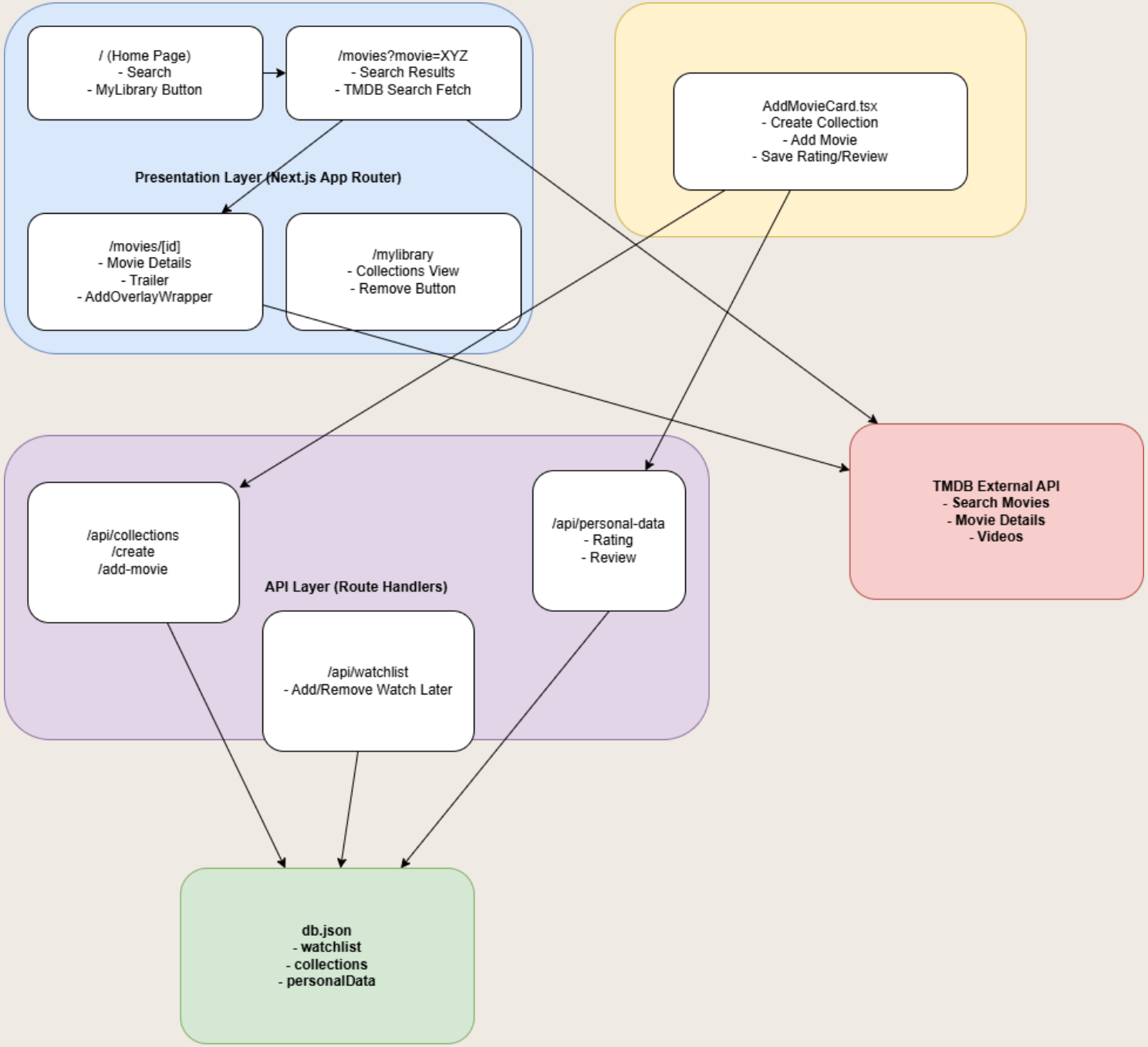
Architecture



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Main Features

01 Real-time movie discovery via a protected Next.js backend.

02 Page generation for unique movie metadata and trailers.

03 Full CRUD functionality to build, name, and manage custom libraries.

04 Automatic retrieval and embedding of official YouTube movie trailers.

05 Merges live TMDB global info with local personal ratings and reviews.

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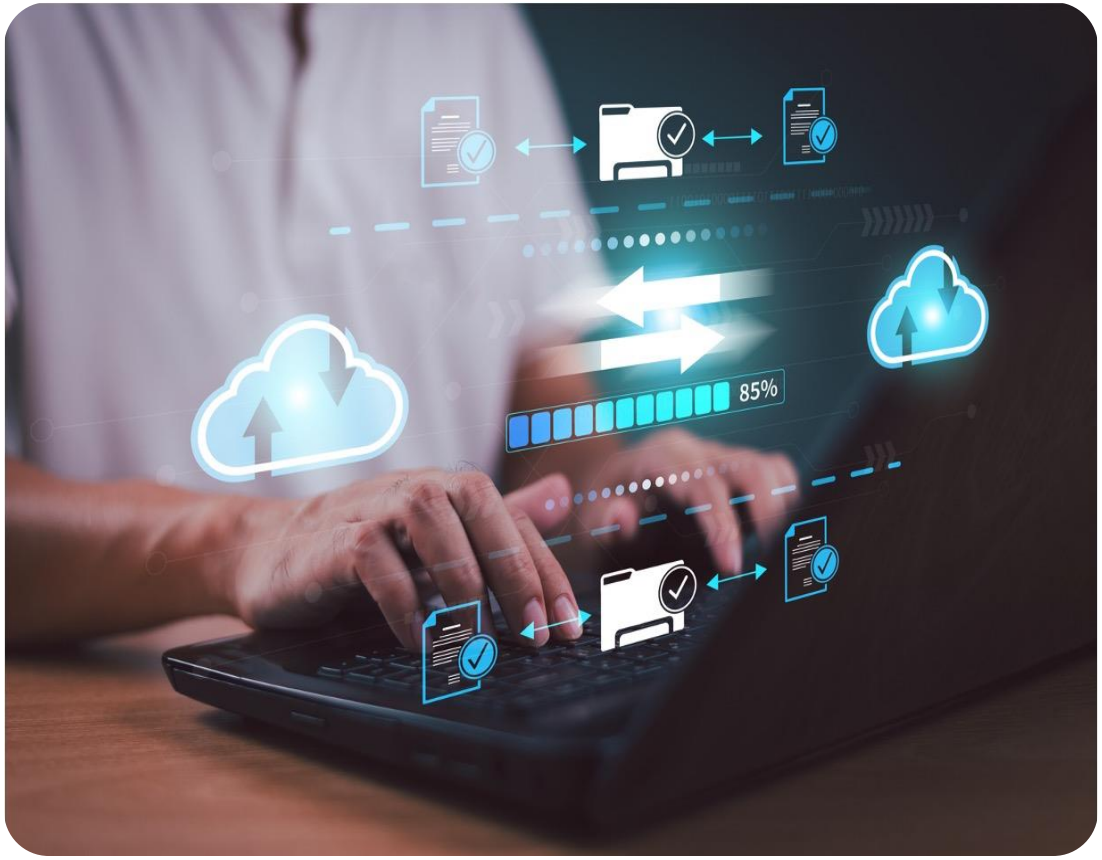
Data Flow

- **1. Client (User):** User enters "Batman" in the search bar. The browser navigates to `/movie?movie=Batman`.
- **2. Frontend (Next.js Page):** The page reads the query and sends a request to the internal backend at `/api/movie`.
- **3. Backend (Route Handler):** The server receives the query, attaches the **API Key**, and fetches data from **TMDB**.
- **4. Local Database (`db.json`):** If the user views their library, the backend reads or writes personal ratings and collections to this file.
- **5. Response:** The backend sends the combined JSON data back to the frontend to render the Movie Cards.



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